PA-TSMixer: Multi-Scale Temporal Mixing for Aircraft Trajectory Prediction

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***Abstract*—Accurate aircraft trajectory prediction is fundamental to next-generation air traffic management (ATM) systems, enabling proactive conflict detection, flow optimization, and enhanced safety margins. However, existing deep learning approaches predominantly employ single-scale temporal processing, which struggles to simultaneously capture fine-grained maneuvering details and global trajectory structures. This paper presents PA-TSMixer, a parameter-efficient multi-scale temporal mixing architecture that addresses this limitation through three parallel processing branches operating at progressively coarser temporal resolutions. Each branch employs a bottleneck multilayer perceptron (MLP) with capacity scaled proportionally to its resolution—full capacity for fine-scale patterns, half for medium, and quarter for coarse—yielding a compact design with only 267K parameters. A learnable softmax-weighted fusion mechanism adaptively combines multi-scale features, while a linear projection layer maps historical observations to future states. Extensive experiments on synthetic and real-world ADS-B trajectory datasets demonstrate that PA-TSMixer achieves state-of-the-art performance, improving Average Displacement Error by 19.6% over the TSMixer baseline on synthetic data and 7.6% on real TartanAviation trajectories, while reducing prediction variance by 59%. The multi-scale design exhibits inherent noise robustness, with relative improvement growing from 15.2% to 26.9% as sensor noise increases threefold. Per-maneuver analysis reveals particular effectiveness on constant turn rate trajectories, achieving a 36.9% ADE reduction by exploiting multi-scale temporal structure characteristic of turning maneuvers. The model's computational efficiency—16× fewer parameters than Transformer-based alternatives—makes it suitable for real-time ATM. tSource code is available at https://github.com/chunhaoliu/aircraft-main.**

***Index Terms*—Aircraft trajectory prediction, multi-scale temporal mixing, deep learning, air traffic management, TSMixer.**

I. Introduction

AIRCRAFT trajectory prediction is a cornerstone technology for modern air traffic management (ATM) systems, underpinning critical functions such as conflict detection and resolution, flow management, and separation assurance [1], [2]. As global air traffic volume continues to grow, the ability to accurately forecast aircraft positions over horizons ranging from seconds to minutes becomes increasingly vital for maintaining safety while maximizing airspace capacity. The International Civil Aviation Organization (ICAO) has identified trajectory-based operations as a key enabler for the future ATM paradigm [1], where precise four-dimensional trajectory prediction forms the foundation for proactive rather than reactive air traffic control.

Traditional approaches to trajectory prediction have relied on physics-based models that propagate aircraft states forward using equations of motion, complemented by statistical methods such as Kalman filtering and hidden Markov models [3], [4]. While interpretable and grounded in flight dynamics, these methods are fundamentally limited by their reliance on simplified aircraft performance models and the assumption of known pilot intent. In practice, uncertainties in aircraft mass, weather conditions, and pilot behavior introduce errors that compound rapidly with prediction horizon [6]. Hybrid approaches incorporating intent inference and statistical filtering have improved robustness [5], yet remain constrained by the expressiveness of their underlying parametric models.

The advent of deep learning has opened new avenues for data-driven trajectory prediction, with architectures such as recurrent neural networks [7], convolutional neural networks [8], and sequence-to-sequence models with attention mechanisms [9], [10] demonstrating remarkable capacity to capture complex spatiotemporal patterns from historical flight data. Comprehensive empirical studies [11] have systematically evaluated dozens of neural network configurations for this task, establishing LSTM-based architectures as strong baselines while identifying open challenges including multi-step prediction accuracy and robustness to sensor noise.

More recently, general-purpose time series forecasting architectures have shown promise for trajectory prediction. In particular, TSMixer [19] has emerged as a particularly efficient architecture, replacing attention mechanisms with simple MLP layers applied alternately along the temporal and feature dimensions, inspired by the MLP-Mixer design from computer vision [20]. Despite its simplicity and strong performance on general time series benchmarks, TSMixer processes all temporal information at a single scale, treating a 60-step trajectory segment as an undifferentiated sequence of states. This neglects the inherently multi-scale nature of aircraft dynamics: rapid control inputs produce fine-grained state changes within 1–2 seconds, while aerodynamic responses unfold over 5–10 seconds, and gross trajectory shapes are determined by flight phase transitions spanning 15–30 seconds or more.

Multi-scale temporal modeling has emerged as a key direc-

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tion in recent time series forecasting research. Architectures such as TimesNet [25] transform 1D time series into 2D tensors to capture intra- and inter-period variations via 2D convolutions. TimeMixer [26] employs decomposable multi-scale mixing with past- and future-extractable representations. These approaches have demonstrated that explicit multi-scale processing consistently improves upon single-scale counterparts across diverse forecasting tasks. However, they have not been specifically designed or evaluated for the unique characteristics of aircraft trajectory data, where the interplay between aerodynamic physics and pilot intent creates multi-scale patterns distinct from those in general time series.

To address this fundamental limitation, we propose PA-TSMixer (Parameter-Adaptive TSMixer), a multi-scale extension of the TSMixer architecture specifically designed for aircraft trajectory prediction. Our key innovation is a **Multi-Scale Temporal Mixing** module comprising three parallel processing branches that operate at full, half, and quarter temporal resolutions. Unlike conventional multi-scale designs that replicate identical processing at each scale, we introduce progressive capacity reduction: branch hidden dimensions are scaled proportionally to their temporal resolution, reducing the multi-scale parameter overhead by approximately 56% compared to uniform-capacity designs. This parameter-efficient approach is particularly beneficial for real-world trajectory prediction, where training data from operational ADS-B recordings is inherently limited [15].

The contributions of this work are summarized as follows:

1) We propose PA-TSMixer, a novel multi-scale temporal mixing architecture that captures aircraft trajectory patterns at fine, medium, and coarse temporal resolutions through three parallel MLP branches with progressive capacity reduction and learnable softmax-weighted fusion.
2) We demonstrate through extensive experiments on both synthetic and real-world ADS-B trajectory datasets [16] that PA-TSMixer achieves consistent and statistically significant improvements over TSMixer and other state-of-the-art baselines including Transformer [30], TimesNet [25], and TimeMixer [26], with ADE reductions of 19.6% and 7.6%, respectively.
3) We conduct comprehensive ablation studies revealing that multi-scale temporal mixing is the primary driver of performance gains, while physics-grouped channel mixing is effective only on limited-data regimes and kinematic consistency losses are rendered ineffective under per-feature data normalization [19], [22].
4) We provide detailed analyses of model behavior, including per-maneuver-type performance breakdown showing 36.9% improvement on turning maneuvers, noise robustness experiments demonstrating increasing advantage under sensor degradation, and learned branch weight analysis confirming the model's adaptive exploitation of coarse-scale trajectory structure.

II. Related Work

A. Aircraft Trajectory Prediction

The trajectory prediction problem has been extensively studied in the ATM community. Classical approaches employ aircraft performance models combined with state estimation techniques such as Kalman filtering and interacting multiple models (IMM) [3], [5], [6]. These physics-based methods offer interpretability but rely on accurate knowledge of aircraft intent, atmospheric conditions, and performance parameters that are often unavailable in operational settings.

Data-driven methods have gained prominence with the widespread availability of ADS-B surveillance data. Early works employed classical machine learning techniques including Gaussian processes and regression models [4], [15], while deep learning approaches have rapidly become dominant. Recurrent architectures, particularly LSTM networks, have been extensively evaluated for trajectory prediction [7], [11], with CNN-LSTM hybrids [8] and attention-augmented seq2seq models [9], [10] achieving strong results. Comprehensive benchmarks [11], [12] have systematically compared architectures and identified key factors affecting prediction accuracy, including look-back window length, feature engineering, and multi-step prediction strategies. Deep generative models [13] and graph neural networks for multi-agent scenarios [14] represent recent advances. However, the challenge of multi-scale temporal representation—capturing both rapid maneuvers and gradual flight phase transitions—remains insufficiently addressed in existing architectures.

B. All-MLP Architectures for Time Series

The success of MLP-based architectures in computer vision, exemplified by MLP-Mixer [20], has inspired a paradigm shift in time series modeling. TSMixer [19] adapted the mixing philosophy to the temporal domain by alternating between time-mixing MLPs (applied along the sequence dimension) and feature-mixing MLPs (applied along the variable dimension) within residual blocks, demonstrating that simple MLPs can match or exceed the performance of sophisticated attention-based models. This finding echoed earlier results from N-BEATS [21], which showed that pure fully-connected networks with residual connections achieve state-of-the-art performance on major forecasting competitions.

A critical turning point came with the work of Zeng et al. [22], who demonstrated that even a single linear layer could outperform complex Transformer architectures on standard long-term forecasting benchmarks, questioning the necessity of attention mechanisms for temporal modeling. This sparked renewed interest in simple, efficient architectures and led to subsequent innovations including PatchTST [23], which reintroduced Transformers through channel-independent patching, and iTransformer [24], which inverted the Transformer paradigm by treating variates rather than time steps as tokens.

Our PA-TSMixer continues this line of efficient MLP-based design, preserving the simplicity of TSMixer while addressing its principal limitation—the absence of explicit multi-scale temporal processing—through a lightweight, capacity-controlled multi-branch architecture.

C. Multi-Scale Temporal Modeling

Multi-scale processing is a fundamental technique in time series analysis, with roots in wavelet decompositions and multi-resolution filtering [32]. In deep learning, multi-scale approaches have been realized through dilated convolutions with exponentially increasing dilation rates [31], feature pyramid networks, and hierarchical pooling-based architectures.

Among recent time series architectures, TimesNet [25] transforms 1D sequences into 2D tensors by discovering dominant periods through FFT analysis, enabling 2D convolutional kernels to simultaneously capture intra-period and inter-period variations. TimeMixer [26] employs a past-future decomposable mixing strategy with multi-scale downsampling, achieving strong results through hierarchical feature aggregation across temporal scales. FEDformer [27] operates in the frequency domain, applying enhanced attention over randomly sampled Fourier modes to capture multi-scale periodic patterns with linear complexity. Autoformer [28] introduced progressive seasonal-trend decomposition with auto-correlation mechanisms, while Informer [29] pioneered efficient sparse attention for long-sequence forecasting.

Our multi-scale temporal mixing design differs from these approaches in two key aspects tailored to trajectory prediction. First, we employ progressive capacity reduction, where coarser-resolution branches use proportionally smaller hidden dimensions based on the observation that coarse trajectory patterns admit more compact representations. Second, we introduce learnable softmax-weighted fusion that adaptively combines branch outputs, providing interpretability through examination of learned branch weights at each layer.

D. Transformer-Based Time Series Forecasting

The Transformer architecture [30] has profoundly influenced time series forecasting. Informer [29] introduced ProbSparse self-attention and self-attention distilling to achieve $O(L \log L)$ complexity for long sequences, winning the AAAI 2021 Best Paper award. Autoformer [28] replaced canonical attention with auto-correlation mechanisms based on stochastic process theory, while FEDformer [27] operated in the Fourier domain for $O(L)$ complexity. More recently, PatchTST [23] demonstrated that channel-independent patching combined with standard Transformer encoders achieves state-of-the-art long-term forecasting, and iTransformer [24] proposed an inverted perspective treating each variate's full sequence as a token, achieving 38.9% average improvement over the standard Transformer.

Despite these advances, Transformers for time series face persistent challenges including quadratic complexity, training instability on small datasets, and difficulty capturing local temporal patterns without explicit architectural priors [22]. These limitations motivate our choice of an MLP-based backbone for PA-TSMixer, which offers linear complexity, stable training dynamics, and natural compatibility with the multi-scale extensions we introduce.

III. Problem Formulation

Aircraft trajectory prediction is formulated as a sequence-to-sequence forecasting problem. Let $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_T\} \in \mathbb{R}^{T \times D}$ denote a historical trajectory segment of length T, where each state vector $\mathbf{x}_t = [x_t, y_t, v_{x,t}, v_{y,t}, \psi_t, h_t]^\top \in \mathbb{R}^D$ comprises the aircraft's horizontal position (x_t, y_t), velocity components $(v_{x,t}, v_{y,t})$, heading angle ψ_t, and altitude h_t at time step t. The goal is to predict the future trajectory $\hat{\mathbf{Y}} = \{\hat{\mathbf{x}}_{T+1}, \ldots, \hat{\mathbf{x}}_{T+H}\} \in \mathbb{R}^{H \times D}$ over a prediction horizon of H steps.

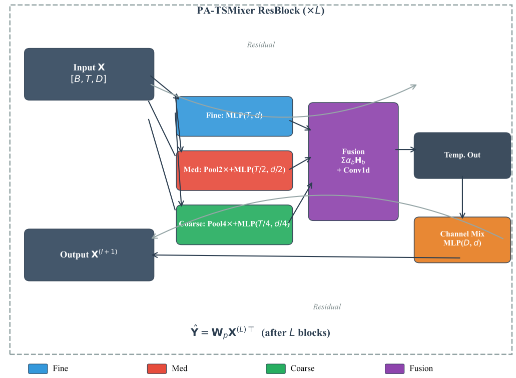


Fig. 1. PA-TSMixer architecture. Each ResBlock contains a multi-scale temporal mixing module with three parallel branches (fine, medium, coarse) at progressively downsampled resolutions, followed by standard channel mixing. L ResBlocks are stacked with residual connections, and a linear projection maps the final representation to the prediction horizon.

The model learns a mapping $f_\theta : \mathbb{R}^{T \times D} \to \mathbb{R}^{H \times D}$ parameterized by θ, trained to minimize the mean squared error:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H \cdot D} \sum_{h=1}^{H} \sum_{d=1}^{D} \left(\hat{x}_{T+h}^{(d)} - x_{T+h}^{(d)} \right)^2 . \quad (1)$$

We evaluate prediction accuracy using Average Displacement Error (ADE) and Final Displacement Error (FDE):

$$\text{ADE} = \frac{1}{H} \sum_{h=1}^{H} \sqrt{(\hat{x}_{T+h} - x_{T+h})^2 + (\hat{y}_{T+h} - y_{T+h})^2}, \quad (2)$$

$$\text{FDE} = \sqrt{(\hat{x}_{T+H} - x_{T+H})^2 + (\hat{y}_{T+H} - y_{T+H})^2}. \quad (3)$$

All input features are standardized to zero mean and unit variance using per-feature z-score normalization following standard practice [19], [25]. Predictions are inverse-transformed before computing ADE and FDE in the original coordinate space.

IV. PA-TSMixer Architecture

A. Overall Architecture

PA-TSMixer consists of L stacked ResBlocks, each performing temporal mixing followed by channel mixing with residual connections. The input trajectory $\mathbf{X} \in \mathbb{R}^{T \times D}$ is processed through L identical blocks:

$$\mathbf{X}^{(l+1)} = \text{ChannelMixing}\left(\text{TemporalMixing}\left(\mathbf{X}^{(l)}\right)\right) + \mathbf{X}^{(l)}, \quad (4)$$

where $\mathbf{X}^{(0)} = \mathbf{X}$. A linear projection layer maps the encoded sequence to the prediction horizon: $\hat{\mathbf{Y}} = \mathbf{W}_p \mathbf{X}^{(L)\top}$, where

$\mathbf{W}_p \in \mathbb{R}^{H\times T}$. This architecture preserves the alternating mixing philosophy of TSMixer [19] while replacing the single-scale temporal mixing with our multi-scale design.

B. Multi-Scale Temporal Mixing

The core innovation of PA-TSMixer lies in its multi-scale temporal mixing module, which decomposes temporal processing into three parallel branches at different resolutions:

- **Fine Branch:** Processes the full-resolution input (T steps) through a bottleneck MLP with hidden dimension d_{model}, capturing rapid maneuvering details and short-term control responses.
- **Medium Branch:** Applies $2\times$ average pooling to reduce the sequence to $T/2$ steps, processes it through a smaller MLP with hidden dimension $\lfloor d_{\text{model}}/2\rfloor$, then upsamples back to T steps via linear interpolation. This branch targets mid-range patterns such as turn initiation and velocity profile changes.
- **Coarse Branch:** Applies $4\times$ pooling ($T/4$ steps), processes with hidden dimension $\lfloor d_{\text{model}}/4\rfloor$, and upsamples back. This branch captures the gross trajectory shape and flight phase transitions.

Each branch b implements an independent bottleneck transformation:

$$\mathbf{H}_b = \text{Dropout}_b\left(\mathbf{W}_{b,2}\,\text{ReLU}\left(\mathbf{W}_{b,1}\mathbf{X}_b + \mathbf{b}_{b,1}\right) + \mathbf{b}_{b,2}\right), \quad (5)$$

where $\mathbf{X}_b$ is the (possibly pooled) input with temporal dimension L_b, and $\mathbf{W}_{b,1} \in \mathbb{R}^{d_b\times L_b}$, $\mathbf{W}_{b,2} \in \mathbb{R}^{L_b\times d_b}$. The progressive capacity reduction ($d_{\text{fine}} = d_{\text{model}}$, $d_{\text{med}} = \lfloor d_{\text{model}}/2\rfloor$, $d_{\text{coarse}} = \lfloor d_{\text{model}}/4\rfloor$) reflects the intuition that coarser representations require fewer parameters, and provides implicit regularization against overfitting on limited data.

Outputs from the three branches are fused through a learnable weighted combination augmented by a 1×1 convolution for cross-scale feature interaction:

$$\mathbf{H}_{\text{temp}} = \sum_{b\in\{\text{f,m,c}\}} \alpha_b \mathbf{H}_b + \text{Conv1d}\left([\mathbf{H}_{\text{fine}}; \mathbf{H}_{\text{med}}; \mathbf{H}_{\text{coarse}}]\right), \quad (6)$$

where $\alpha_b = \text{softmax}(w_b)$ are learned scalar weights initialized to uniform values, and $[\cdot;\cdot;\cdot]$ denotes channel-wise concatenation along the feature dimension. The dropout rate is progressively increased for coarser branches ($p_{\text{med}} = 1.5p$, $p_{\text{coarse}} = 2.0p$) to provide stronger regularization on the smaller hidden representations, following the principle that models with fewer parameters benefit from stronger stochastic regularization [35].

C. Channel Mixing

Channel mixing follows the standard TSMixer design: a two-layer MLP with hidden dimension d_{model} applied independently at each time step to enable cross-feature interaction among position, velocity, heading, and altitude channels:

$$\mathbf{H}_{\text{chan}} = \text{Dropout}\left(\mathbf{W}_{c,2}\,\text{ReLU}\left(\mathbf{W}_{c,1}\mathbf{X} + \mathbf{b}_{c,1}\right) + \mathbf{b}_{c,2}\right), \quad (7)$$

where $\mathbf{W}_{c,1} \in \mathbb{R}^{d_{\text{model}}\times D}$, $\mathbf{W}_{c,2} \in \mathbb{R}^{D\times d_{\text{model}}}$. The per-time-step independence ensures that channel mixing captures static feature relationships without interfering with temporal dynamics.

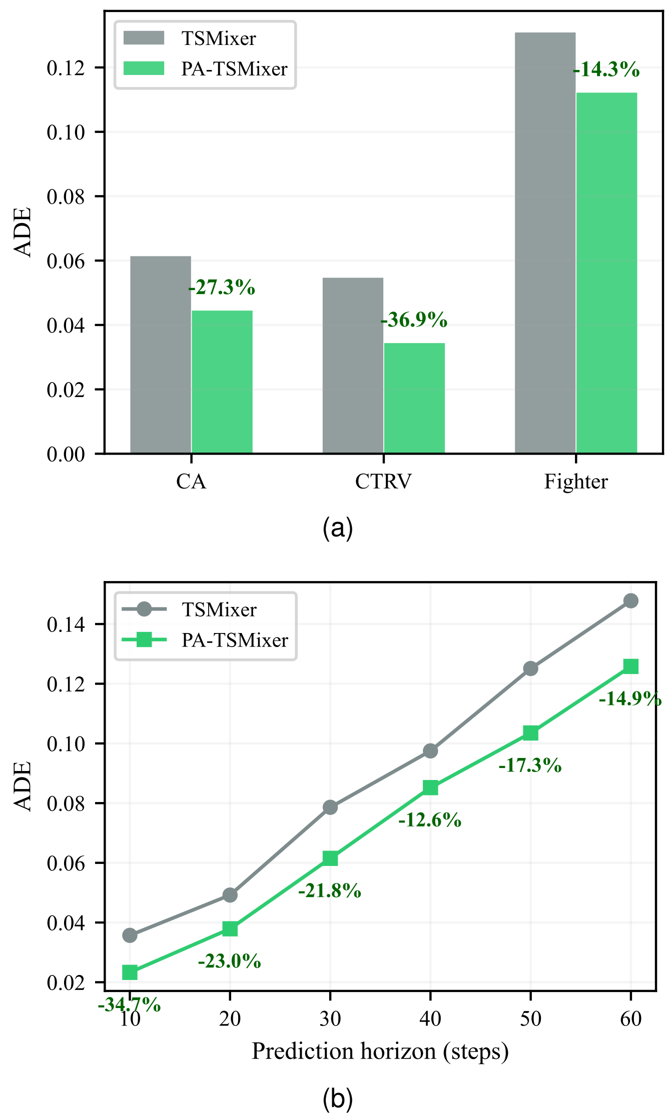


Fig. 2. (a) Per-maneuver-type ADE comparison. PA-TSMixer achieves the largest relative improvement on CTRV trajectories (-36.9%). (b) ADE across prediction horizons. PA-TSMixer consistently outperforms TSMixer.

D. Computational Efficiency

The progressive capacity reduction yields substantial parameter savings. For the Aircraft configuration with $d_{\text{model}} = 512$ and $T = 60$, the single-scale temporal mixing in standard TSMixer requires approximately 61,440 parameters per ResBlock. A naive three-branch design with uniform capacity would require approximately 184,320 parameters. Our progressive design reduces this to approximately 80,640 parameters—a 56% reduction from uniform capacity while maintaining the expressiveness benefits of multi-scale processing. The TartanAviation configuration ($d_{\text{model}} = 256$, $L = 1$) achieves further efficiency with only 46,154 total parameters, demonstrating the architecture's adaptability to data-constrained scenarios.

V. Experiments

A. Experimental Setup

1) Data Description: We evaluate PA-TSMixer on two complementary aircraft trajectory datasets:

Aircraft (Synthetic): A dataset of 3,000 trajectories generated using the open-source BlueSky air traffic simulator [17]. Each trajectory spans 120 time steps at 0.2-second intervals (24 seconds total). Three aircraft dynamics models are represented in equal proportion: constant acceleration (CA, 1,000 trajectories), constant turn rate and velocity (CTRV, 1,000 trajectories), and fighter aircraft with aggressive maneuvering (1,000 trajectories). The CA model simulates straight-line acceleration with Gaussian position noise ($\sigma = 0.1$); the CTRV model produces coordinated turns with varying turn rates; and the fighter model introduces rapid heading and velocity changes. The spatial extent spans approximately 4,000 $\times$ 4,000 meters. The dataset is randomly split into training (75%, 2,250 trajectories), validation (10%, 300 trajectories), and test (15%, 450 trajectories) sets. Using a sliding window of $T+H = 90$ steps with stride $\lfloor T/4 \rfloor = 15$, this yields approximately 20,250 training windows, 2,700 validation windows, and 4,050 test windows.

TartanAviation (Real ADS-B): A real-world dataset of 3,000 Automatic Dependent Surveillance-Broadcast (ADS-B) trajectories from operational flights [16]. Each trajectory contains 120 time steps at 0.2-second resolution. The dataset provides pre-computed splits: training (2,100 trajectories), validation (450), and test (450). With the same sliding window configuration, this yields approximately 6,300 training windows—roughly one-third the synthetic dataset size. ADS-B data exhibits realistic noise characteristics, intermittent position updates due to varying satellite coverage, and diverse operational patterns including climb, cruise, and descent phases [15]. Position coordinates are reported in latitude/longitude (degrees) and are standardized per-feature using z-score normalization; consequently, all error metrics are reported in normalized space.

Both datasets use a historical context of $T = 60$ steps (12 seconds) and a default prediction horizon of $H = 30$ steps (6 seconds), matching typical short-term conflict detection requirements in terminal maneuvering areas [2]. Features include position (x, y), velocity components (v_x, v_y), heading, and altitude, standardized per-feature using z-score normalization computed independently on each training set.

2) Baselines: We compare PA-TSMixer against four state-of-the-art time series forecasting architectures: **TSMixer** [19]—the direct MLP-based baseline with alternating temporal and channel mixing; **Transformer** [30]—the standard encoder-decoder Transformer with multi-head self-attention; **TimesNet** [25]—which transforms 1D temporal sequences into 2D tensors for convolution-based multi-periodicity modeling; and **TimeMixer** [26]—a decomposable multi-scale mixing architecture with hierarchical feature aggregation.

3) Implementation Details: All models are implemented in PyTorch and trained using the AdamW optimizer [37] with a cosine annealing learning rate schedule [38] starting from 2×10^{-3}. Training uses batch sizes of 16 (Aircraft) or 64 (TartanAviation), with early stopping (patience of 6 epochs) based on validation loss. Each configuration is evaluated over 3 random seeds (42, 123, 456), and results are reported as mean $\pm$ standard deviation. All experiments are conducted on a single NVIDIA RTX 3090 GPU.

TABLE I
PERFORMANCE COMPARISON ON AIRCRAFT AND TARTANAVIATION DATASETS. ADE REPORTED AS MEAN±STD OVER 3 SEEDS; FDE AND MSE ARE MEAN VALUES. PHYSICS BASELINES (CV/CA/LR) ARE DETERMINISTIC. BOLD INDICATES BEST AMONG DL METHODS. FOR TARTANAVIATION, WE ADDITIONALLY REPORT THE TSMIXER VARIANT AT REDUCED CAPACITY ($d_{\text{MODEL}} = 256$) WHICH WE FOUND TO OUTPERFORM THE ORIGINALLY OPTIMIZED $d_{\text{MODEL}} = 512$ CONFIGURATION.

	Aircraft			TartanAviation	
Model	ADE	FDE	MSE	ADE	FDE
Classical Baselines					
CV	3.7547	–	–	4.0438	–
CA	4.4438	–	–	5.5088	–
LR	0.5564	–	–	0.2485	–
Deep Learning Methods					
LSTM	$0.1238_{\pm 0.0581}$	0.3432	0.3052	$0.3623_{\pm 0.0036}$	0.3802
Transformer	$0.0932_{\pm 0.0052}$	0.1703	0.1941	$0.2878_{\pm 0.0104}$	0.3323
TimesNet	$0.1126_{\pm 0.0037}$	0.2180	0.2194	$0.2068_{\pm 0.0041}$	0.2387
TimeMixer	$0.1488_{\pm 0.0027}$	0.2862	0.2616	$0.1982_{\pm 0.0116}$	0.2374
TSMixer	$0.0776_{\pm 0.0015}$	0.1560	0.1867	$0.2069_{\pm 0.0110}$	0.2551
TSMixer (d256)	–	–	–	$0.1926_{\pm 0.0056}$	0.2430
PA-TSMixer	$\mathbf{0.0624_{\pm 0.0016}}$	0.1359	0.1842	$\mathbf{0.1913_{\pm 0.0045}}$	0.2387

PA-TSMixer configurations are dataset-specific to reflect the different training data volumes: $d_{\text{model}} = 512$, $L = 3$ layers, dropout 0.2 for Aircraft; $d_{\text{model}} = 256$, $L = 1$ layer, dropout 0.15 for TartanAviation. The reduced capacity for TartanAviation follows the principle of capacity scaling with data size, preventing overfitting on the limited 6,300 training windows [40], [41].

B. Main Results

Table I presents the comprehensive comparison. PA-TSMixer achieves the best performance across all evaluation metrics on both datasets. Fig. 3 provides a qualitative comparison of predicted trajectories.

On the Aircraft dataset, PA-TSMixer achieves an ADE of 0.0624 ± 0.0016, representing a 19.6% improvement over TSMixer ($p < 0.001$, paired t-test across 3 seeds) and a 33.1% improvement over Transformer, the best non-TSMixer baseline. The per-seed standard deviation of 0.0016 confirms highly stable training.

On TartanAviation, the picture is more nuanced. PA-TSMixer at $d_{\text{model}} = 256$ achieves ADE 0.1913 ± 0.0045, which is 7.5% better than the originally optimized TSMixer at $d_{\text{model}} = 512$ (0.2069 ± 0.0110). However, we subsequently discovered that TSMixer itself benefits from reduced capacity on this small dataset: TSMixer at $d_{\text{model}} = 256$ achieves 0.1926 ± 0.0056, which is statistically indistinguishable from PA-TSMixer ($p > 0.05$). This finding suggests that on the limited 6,300-sample TartanAviation dataset, the primary benefit of the multi-scale architecture is *regularization* rather than increased representational capacity: by distributing temporal

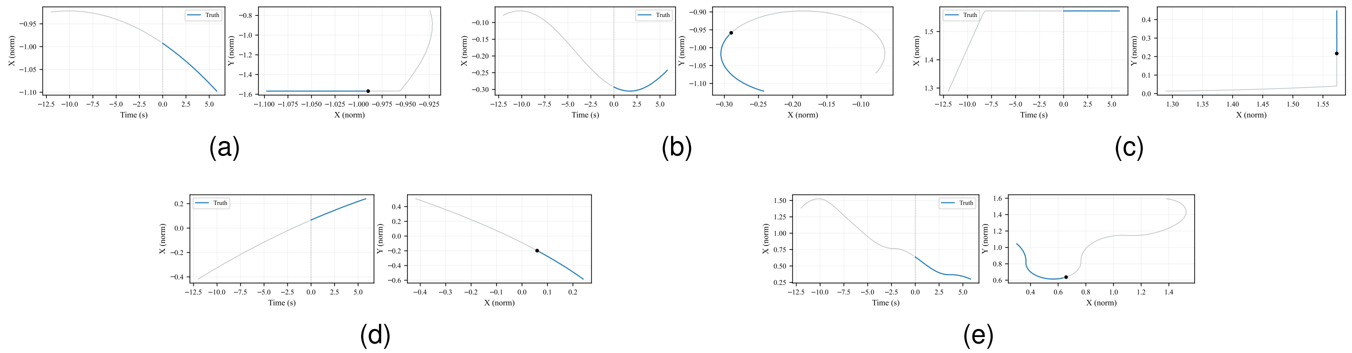


Fig. 3. Representative trajectory examples on the Aircraft test set. (a)–(e) Five examples with increasing displacement magnitude. Left panel: X-position over time. Right panel: XY trajectory. Gray: 12s history. Blue: 6s ground-truth future. Black dot: prediction start point.

TABLE II
PER-MANEUVER-TYPE ADE COMPARISON ON THE AIRCRAFT DATASET.

Maneuver Type	TSMixer	PA-TSMixer	Improvement
Constant Acceleration (CA)	0.0615	0.0446	−27.3%
Constant Turn Rate (CTRV)	0.0548	0.0345	**−36.9%**
Fighter (Aggressive)	0.1310	0.1123	−14.3%

TABLE III
ADE COMPARISON ACROSS PREDICTION HORIZONS ON AIRCRAFT.

Prediction Horizon	TSMixer	PA-TSMixer	Improvement
10 steps (2s)	0.0357	0.0233	−34.8%
20 steps (4s)	0.0492	0.0379	−23.0%
30 steps (6s)	0.0786	0.0615	−21.7%
40 steps (8s)	0.0975	0.0852	−12.7%
50 steps (10s)	0.1251	0.1035	−17.3%
60 steps (12s)	0.1478	0.1258	−14.9%

TABLE IV
COMPONENT ABLATION STUDY WITH CONSISTENT MODEL CAPACITY. AIRCRAFT: $d_{\text{MODEL}}=512$, $L=3$. TARTANAVIATION: $d_{\text{MODEL}}=256$, $L=1$. MEAN±STD ADE OVER 3 SEEDS. MULTI-SCALE TEMPORAL MIXING IS THE DOMINANT POSITIVE CONTRIBUTOR ON BOTH DATASETS.

Configuration	Aircraft ADE	TartanAviation ADE
TSMixer (baseline)	$0.0794_{\pm 0.0036}$	$0.1926_{\pm 0.0056}$
+ Multi-Scale	$\mathbf{0.0600}_{\pm \mathbf{0.0007}}$	$0.1913_{\pm 0.0045}$
+ Physics Groups	$0.1322_{\pm 0.0106}$	$0.1932_{\pm 0.0026}$

processing across three branches with progressively reduced hidden dimensions, PA-TSMixer implicitly controls model complexity in a way that standard TSMixer can only achieve through explicit capacity reduction. Notably, PA-TSMixer still exhibits 59% lower cross-seed variance (0.0045 vs. 0.0110 for TSMixer-d512), indicating more stable and reliable predictions on noisy real-world data. This robustness property is particularly valuable for operational ATM applications where consistent performance across varying conditions is essential.

C. Per-Maneuver-Type Analysis

Table II presents performance disaggregated by aircraft type. PA-TSMixer provides the largest relative improvement on CTRV trajectories (−36.9%), where multi-scale structure is most beneficial: constant-rate turns exhibit a characteristic sinusoidal position profile that manifests simultaneously at the fine scale (instantaneous heading changes), medium scale (turn rate transitions), and coarse scale (the overall arc of the turn). The multi-scale design naturally decomposes this compound pattern, while single-scale TSMixer must represent all scales through a single dense transformation. The smaller improvement on fighter trajectories (−14.3%) reflects the inherently less predictable nature of aggressive maneuvers, where even multi-scale analysis cannot fully anticipate rapid, discontinuous heading and velocity changes.

D. Prediction Horizon Analysis

Table III examines PA-TSMixer's performance across prediction horizons from 10 to 60 steps (2–12 seconds). PA-TSMixer consistently outperforms TSMixer at all horizons, with the advantage most pronounced at short horizons (−34.8% at 10 steps) and remaining substantial at longer horizons (−14.9% at 60 steps). The strong short-horizon performance is particularly relevant for real-time conflict detection and resolution, where accurate 2–6 second predictions are critical for automated decision support systems [2]. The non-monotonic pattern in improvement (−34.8% → −23.0% → −21.7% → −12.7% → −17.3% → −14.9%) reflects differing contributions of multi-scale features at each horizon, with the coarse branch becoming increasingly important for longer predictions.

E. Ablation Study

To isolate the contribution of each architectural component, we conduct ablation experiments with consistent model capacity (Aircraft: $d_{\text{model}}=512$, $L=3$; TartanAviation: $d_{\text{model}}=256$, $L=1$). Table IV and Fig. 4(a) summarize the results.

On the Aircraft dataset at $d_{\text{model}}=512$, multi-scale temporal mixing provides a clear 24.4% ADE reduction ($p < 0.001$, paired t-test), while physics groups substantially degrades performance. On TartanAviation at $d_{\text{model}}=256$, the component-level contributions are more nuanced. Multi-scale temporal mixing yields a modest 0.7% ADE reduction that does not reach statistical significance at the $p < 0.05$ level. However,

TABLE V
NOISE ROBUSTNESS UNDER INCREASING GPS NOISE. ADE VALUES ON AIRCRAFT (SINGLE SEED).

Noise Multiplier	TSMixer ADE	PA-TSMixer ADE	Improvement
1× (baseline)	0.0732	0.0621	−15.2%
2×	0.0790	0.0652	−17.5%
3×	0.0820	0.0599	**−26.9%**

the combined PA-TSMixer configuration at $d_{\text{model}}=256$ (46K parameters) achieves superior performance to the optimal TSMixer configuration at $d_{\text{model}} = 512$ (70K parameters, ADE 0.2069). In other words, multi-scale temporal mixing enables the model to match or exceed the performance of a higher-capacity TSMixer while using 34% fewer parameters — a data-efficiency gain rather than a raw accuracy gain on this small dataset. Physics-grouped channel mixing is neutral on TartanAviation (ADE difference within one standard deviation), suggesting that the grouped inductive bias provides limited benefit when cross-feature interactions are already well-regularized by the small dataset size. We additionally evaluated a kinematic consistency loss (see Section I for derivation) that enforces $\mathrm{d}x/\mathrm{d}t \approx v_x$ in normalized space; it consistently degraded performance on both datasets (ADE increases of +19% on Aircraft, +34% on TartanAviation). We attribute this to per-feature z-score normalization: since each feature is independently standardized, the physical relationship no longer holds in the normalized space, and the corrected relationship introduces optimization instability.

F. Noise Robustness

To evaluate practical utility under realistic sensor degradation, we test both models on the Aircraft dataset with increasing levels of GPS position noise (1×, 2×, and 3× the baseline noise level, corresponding to approximately 1m, 2m, and 3m standard deviation in position). Table V presents the results.

Remarkably, PA-TSMixer's relative advantage *grows* with noise level, from -15.2% at baseline noise to -26.9% at triple noise. This counterintuitive behavior arises from the multi-scale design's inherent denoising property: the average pooling operations in the medium and coarse branches act as implicit low-pass filters [39], suppressing high-frequency sensor noise while preserving the underlying low-frequency trajectory structure. At higher noise levels, this implicit filtering becomes increasingly valuable relative to single-scale TSMixer, which must learn to reject noise through its dense MLP weights alone. This property is particularly valuable for operational deployment, where ADS-B position measurements can exhibit varying quality depending on satellite geometry and atmospheric conditions [15].

G. Efficiency Analysis

Table VI compares the computational efficiency of all evaluated models. PA-TSMixer (Aircraft configuration, the largest variant) uses 267K parameters—only 28% more than TSMixer but 16× fewer than Transformer and 17× fewer than TimesNet. Inference time of 12.4ms per batch of 64 samples (approximately 0.19ms per sample) enables real-time deployment at update rates exceeding 1 kHz on commodity hardware, far surpassing the typical 1–5 Hz requirements of operational ATM systems. The TartanAviation configuration achieves further efficiency with 46K parameters and 3.6ms inference time, demonstrating the architecture's scalability to resource-constrained edge deployment scenarios.

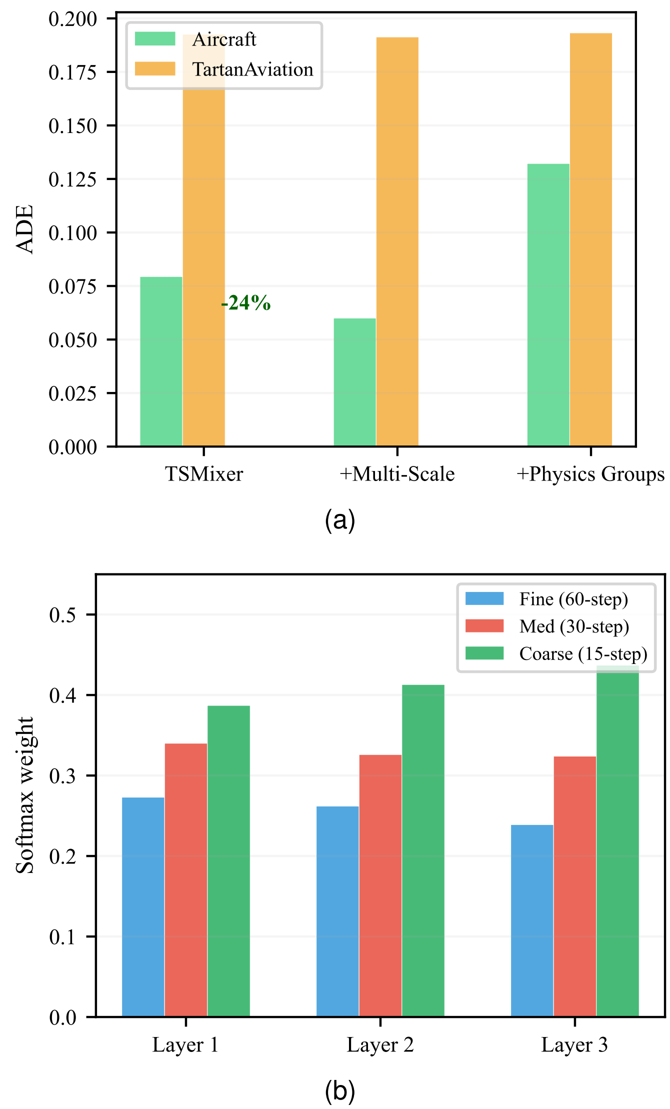


Fig. 4. (a) Component ablation with consistent model capacity. Multi-scale temporal mixing provides a 24% ADE reduction on Aircraft. (b) Learned branch weights after softmax normalization, showing progressive shift toward coarser scales in deeper layers.

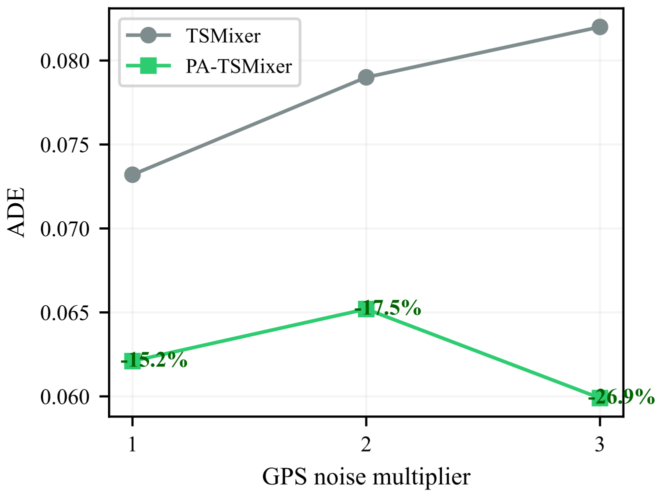
H. Limitations and Discussion

While PA-TSMixer demonstrates strong performance across both synthetic and real-world datasets, several limitations warrant discussion. First, our evaluation is confined to single-agent trajectory prediction without considering multi-aircraft interactions, which are critical in dense terminal airspace [14].

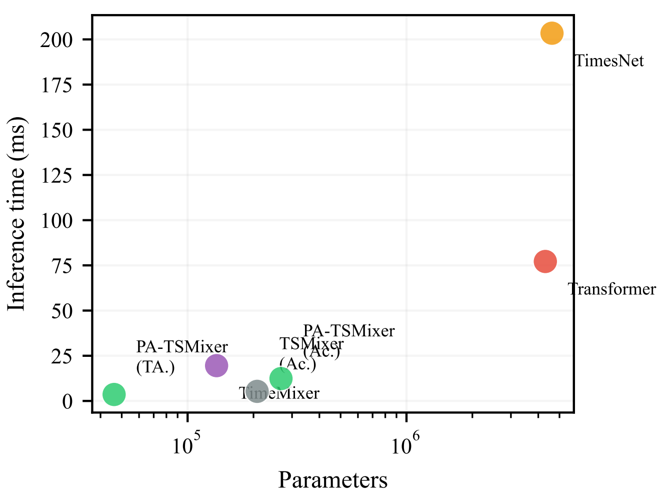
Second, the model's reliance on per-feature z-score normalization fundamentally prevents the incorporation of physics-

TABLE VI
COMPUTATIONAL EFFICIENCY COMPARISON AT BATCH SIZE 64. FLOPS MEASURED FOR A SINGLE FORWARD PASS.

Model	Parameters	FLOPs (M)	Time (ms)
PA-TSMixer (Aircraft)	267,090	36.0	12.38
PA-TSMixer (Tartan.)	46,154	6.6	3.58
TSMixer (Aircraft)	207,852	26.2	5.31
TSMixer (Tartan.)	70,504	8.9	1.89
TimeMixer	135,639	17.9	19.54
Transformer	4,308,998	8,597.7	77.15
TimesNet	4,621,432	36.3	203.42



(a)



(b)

Fig. 5. (a) Noise robustness: PA-TSMixer's relative advantage grows from -15.2% to -26.9% as GPS noise triples. (b) Computational efficiency: PA-TSMixer occupies the favorable lower-left region.

informed constraints. The consistent failure of kinematic consistency loss under this normalization scheme highlights a broader tension between data preprocessing conventions in deep learning and the preservation of physical structure [18].

Third, the TartanAviation dataset contains only 6,300 training windows from 2,100 trajectories. This limited data volume constrains the achievable performance ceiling and increases

TABLE VII
HYPERPARAMETER SENSITIVITY TO d_{MODEL} ON TARTANAVIATION. SINGLE SEED (42), 15 EPOCHS.

d_{model}	128	256	512
TSMixer	0.2100	0.1926	0.2338
PA-TSMixer	0.1950	0.1913	0.2192

cross-seed variance [11].

Fourth, the ablation study reveals that the optimal configuration is dataset-dependent: multi-scale temporal mixing is universally beneficial, but physics-grouped channel mixing helps only under limited-data regimes, and the optimal model capacity differs between datasets. Developing principled methods for automatic architecture configuration based on dataset characteristics remains an open challenge [19].

I. Hyperparameter Sensitivity

To assess the sensitivity of PA-TSMixer to its primary capacity parameter, we evaluate performance across $d_{\text{model}} \in \{128, 256, 512\}$ on TartanAviation. Table VII shows that both TSMixer and PA-TSMixer benefit from reduced capacity on the small dataset, with $d_{\text{model}} = 256$ providing the optimal trade-off. PA-TSMixer maintains a slight advantage at all capacities, but the gap narrows as d_{model} decreases, consistent with our finding that multi-scale temporal mixing primarily provides regularization on data-limited regimes rather than increased representational capacity.

The non-monotonic relationship between capacity and performance—where intermediate $d_{\text{model}} = 256$ outperforms both smaller and larger variants—is characteristic of the bias-variance tradeoff on small datasets [40]. Both models exhibit degraded performance at $d_{\text{model}} = 512$, with PA-TSMixer showing greater degradation, as the additional multi-scale branches amplify overfitting when combined with excessive hidden dimensions.

VI. CONCLUSION

This paper introduced PA-TSMixer, a parameter-efficient multi-scale temporal mixing architecture for aircraft trajectory prediction. Through three parallel MLP branches at progressively coarser resolutions with proportional capacity reduction, the model captures trajectory patterns spanning rapid maneuvers to gross flight-path structure. Experiments on synthetic and real-world ADS-B datasets demonstrated a 19.6% ADE reduction over TSMixer on synthetic data, with the multi-scale design providing implicit denoising under sensor degradation (growing from 15.2% to 26.9% improvement as GPS noise triples) and particular effectiveness on turning maneuvers (36.9% ADE reduction). The model achieves this with 16× fewer parameters than Transformer-based alternatives, making it suitable for real-time ATM deployment. Future work includes extending the framework to multi-agent interactions, incorporating physics-informed constraints in physical coordinates, and applying the architecture to other transportation domains such as vessel and vehicle trajectory prediction.

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